Lesson 1: What is Pygame?

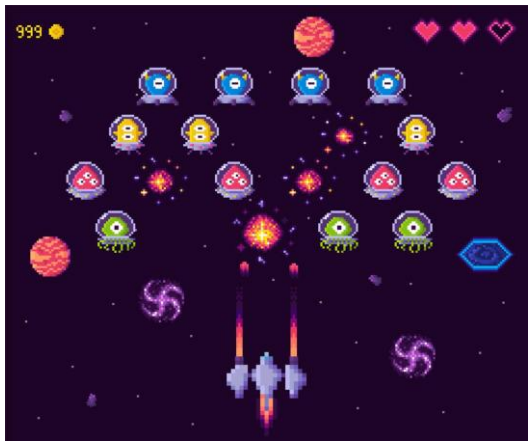
Pygame is a Python library used to create 2D games.

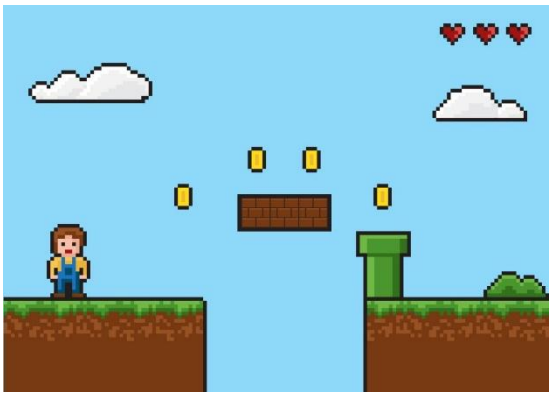
With Pygame, you can create:



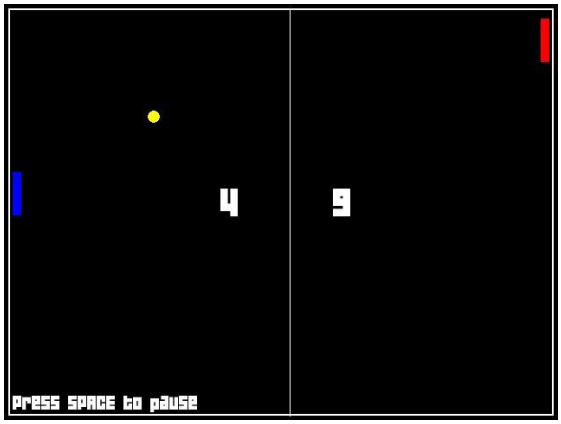
Car Racing

Space Shooter





Platformer



Step 1: Install Pygame

Open your terminal or Command Prompt and type:

```
pip install pygame
```

Step 2: Create your first game window

Create a Python file named `lesson1.py` and write this code:

lesson1.py > ...

```python

import pygame

# Start Pygame
pygame.init()

# Create a window
screen = pygame.display.set_mode((800, 600))

# Set window title
pygame.display.set_caption("My First Game")

# Game loop
running = True

while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    # Background color
    screen.fill((30, 30, 60))

    # Update the display
    pygame.display.flip()

# Close Pygame
pygame.quit()

```

Run it:

```
python lesson1.py
```

Congratulations! 🎉 You have created your first game window.

Step 3: Understand the code

Let's understand the important lines.

Code	Simple meaning
`import pygame`	Bring Pygame into our program.
`pygame.init()`	Start the Pygame modules.
`set_mode((800, 600))`	Create a window 800 pixels wide and 600 pixels high.
`set_caption()`	Give the window a title.
`while running:`	Keep the game running.
`pygame.event.get()`	Check events like key presses and closing the window.
`screen.fill()`	Paint the background.
`pygame.display.flip()`	Show the latest drawing on the screen.
`pygame.quit()`	Close Pygame properly.

What is a game loop?

A game loop is a block of code that runs again and again while the game is active.

For example:

1. Check what the player does.
2. Move the player.
3. Draw the game.
4. Repeat!

Without a game loop, your game cannot continuously respond to the player.

Step 4: Change the background color

Find this line:

```
screen.fill((30, 30, 60))
```

Change it to:

```
screen.fill((0, 150, 0))
```

Run the game again. Your background will become green.

Colors in Pygame use RGB values:

- Red: `(255, 0, 0)`
- Green: `(0, 255, 0)`
- Blue: `(0, 0, 255)`
- White: `(255, 255, 255)`
- Black: `(0, 0, 0)`

Each value ranges from 0 to 255.

Step 5: Your first practical challenge

Pygame Challenge

Change your program to complete these tasks:

- [ ] Make the background red.
- [ ] Change the window size to 1000 × 700.
- [ ] Change the window title to My Racing Game.